

# Assignment

Question 2a)

<https://youtu.be/fpgFEqAN-ls>

Question 3)

<https://youtu.be/n9vg6bFQWxQ>

Question 2b)

<https://youtu.be/5Bokzdg5CCs>

Question 4)

## Example 7.3

Consider the CMOS common-source amplifier in Fig. 7.4(a) for the case  $V_{DD} = 3 \text{ V}$ ,  $V_{tn} = |V_{tp}| = 0.6 \text{ V}$ ,  $\mu_n C_{ox} = 200 \text{ } \mu\text{A/V}^2$ , and  $\mu_p C_{ox} = 65 \text{ } \mu\text{A/V}^2$ . For all transistors,  $L = 0.4 \text{ } \mu\text{m}$  and  $W = 4 \text{ } \mu\text{m}$ . Also,  $V_{An} = 20 \text{ V}$ ,  $|V_{Ap}| = 10 \text{ V}$ , and  $I_{REF} = 100 \text{ } \mu\text{A}$ . Find the small-signal voltage gain. Also, find the coordinates of the extremities of the amplifier region of the transfer characteristic—that is, points A and B.

**Solution**

$$\begin{aligned} g_{m1} &= \sqrt{2k'_n \left(\frac{W}{L}\right)_1 I_{REF}} \\ &= \sqrt{2 \times 200 \times \frac{4}{0.4} \times 100} = 0.63 \text{ mA/V} \end{aligned}$$

$$r_{o1} = \frac{V_{An}}{I_{D1}} = \frac{20 \text{ V}}{0.1 \text{ mA}} = 200 \text{ k}\Omega$$

$$r_{o2} = \frac{|V_{Ap}|}{I_{D2}} = \frac{10 \text{ V}}{0.1 \text{ mA}} = 100 \text{ k}\Omega$$

Thus,

$$\begin{aligned} A_v &= -g_{m1}(r_{o1} \parallel r_{o2}) \\ &= -0.63(\text{mA/V}) \times (200 \parallel 100)(\text{k}\Omega) = -42 \text{ V/V} \end{aligned}$$